ENAE464 - 0202

Lab04: Particle Imaging Velocimetry

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Enclosed is the technical report for the *Particle Imaging Velocimetry* laboratory experiment. This report presents the experimental methodology, results, and analysis of fluid characteristics and imaging effects for flow downstream of a circular cylinder.
Please feel free to contact us with any questions regarding the contents of this report.

Respectfully,
Mikołaj Kostrzewa & Vai Srivastava

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1 Abstract

This report presents an experimental investigation of choked flow in a convergent-divergent duct using the Armfield C1-MkII Compressible Flow Bench. Air was drawn through the duct by a variable-speed compressor, and three pressure taps at the inlet, throat, and outlet were monitored as the outlet pressure was progressively reduced. The mass flow rate $\dot{m}$ and throat pressure ratio $p_{\text{throat}}/p_{\text{atmos}}$ were computed at each operating point and compared against isentropic flow theory. The results clearly demonstrate the onset of choking at an outlet pressure drop of approximately 8 kPa to 9 kPa, beyond which $\dot{m}$ plateaus at 16.76 g/s. This experimentally observed maximum agrees with the theoretical prediction of 17.09 g/s to within 2%. The limiting throat pressure ratio of 0.481 falls 8.9% below the theoretical critical value of $r_c = 0.528$, a discrepancy attributed to viscous losses and boundary layer effects not captured by the inviscid isentropic model.

2 Introduction

The phenomenon of *choking* describes a condition in compressible duct flow where the mass flow rate reaches a maximum and can no longer increase, regardless of further reductions in downstream pressure. This experiment investigates choking in a convergent-divergent duct using the Armfield C1-MkII Compressible Flow Bench, with the objective of demonstrating the onset of choking and comparing experimental results to isentropic flow theory.

2.1 Theoretical Background

For frictionless, adiabatic (isentropic) flow of an ideal gas with no shaft work, the energy equation reduces to:

$$\frac{\gamma}{\gamma - 1}\frac{p}{\rho} + \frac{V^2}{2} = \text{constant} \tag{2.1.1}$$

where $\gamma = c_p/c_v$ is the ratio of specific heats ($\gamma = 1.4$ for air), p is the static pressure, ρ is the density, and V is the flow velocity. Applying Equation 2.1.1 between the atmosphere and the throat, and using the isentropic relation $p/\rho^\gamma = \text{const}$, the mass flow rate through the throat can be expressed as:

$$\dot{m} = \rho_{\text{atmos}}\, a_{\text{throat}} \sqrt{\frac{2\gamma}{\gamma - 1}\frac{p_{\text{atmos}}}{\rho_{\text{atmos}}}\left[r^{2/\gamma} - r^{(\gamma+1)/\gamma}\right]} \tag{2.1.2}$$

where a_{throat} is the throat cross-sectional area and $r = p_{\text{throat}}/p_{\text{atmos}}$ is the ratio of throat pressure to atmospheric pressure.

Inspection of Equation 2.1.2 shows that $\dot{m} = 0$ at both $r = 0$ and $r = 1$, implying a maximum exists for some critical value r_c. Setting $d\dot{m}/dr = 0$ yields:

$$r_c = \left(\frac{2}{\gamma + 1}\right)^{\gamma/(\gamma-1)} \tag{2.1.3}$$

which evaluates to $r_c \approx 0.528$ for air. At this critical pressure ratio, the flow velocity at the throat equals the local speed of sound, i.e. the throat Mach number reaches unity ($M_{\text{throat}} = 1$).

Once $M_{\text{throat}} = 1$, pressure disturbances originating downstream can no longer propagate upstream through the throat, since they travel at the speed of sound. Consequently, further reductions in the outlet pressure p_{outlet} have no effect on the throat conditions: p_{throat} remains fixed at $r_c \cdot p_{\text{atmos}}$, and $\dot{m}$ remains at its maximum value. This condition is referred to as *choked flow*.

2.2 Objectives

The specific objectives of this experiment are:

1. To measure the mass flow rate $\dot{m}$ through the convergent-divergent duct as the outlet pressure is progressively reduced, and to identify the point at which choking first occurs.
2. To observe the behaviour of the throat pressure p_{throat} relative to p_{atmos} before and after choking.
3. To compare the experimentally obtained limiting values of $\dot{m}$ and $p_{\text{throat}}/p_{\text{atmos}}$ against the theoretical predictions of isentropic flow theory, particularly the critical pressure ratio $r_c \approx 0.528$.

3 Methodology

3.1 Experimental Apparatus

The experiment was conducted using the Armfield C1-MKII Compressible Flow Unit, shown in Figure 3.1.1. The apparatus consists of two primary components: a variable-speed motor-driven compressor, and a convergent-divergent nozzle test section fabricated from clear acrylic.

Figure 3.1.1: Converging-Diverging Duct Test Seciton on the C1-MkII Compressible Flow Bench

The convergent-divergent duct features three instrumented cross-sections: the inlet, the throat, and the outlet, with diameters $d_1 = d_3 = 24\,\text{mm}$ and $d_2 = 9.5\,\text{mm}$ respectively. The compressor draws air through the duct from right to left (inlet to outlet), generating a controlled low-pressure region downstream. Three pressure taps connect the duct to a digital control console, which displays the following gauge pressure differences relative to atmospheric pressure:

- $P_1 = p_{\text{atmos}} - p_{\text{throat}}$ [kPa]
- $P_2 = p_{\text{atmos}} - p_{\text{outlet}}$ [kPa]
- $P_3 = p_{\text{atmos}} - p_{\text{inlet}}$ [Pa]

Ambient laboratory temperature and pressure were measured prior to the experiment using a thermometer and barometer, and are recorded in Table 8.1.1.

3.2 Operating Procedure

Prior to starting the compressor, the ambient temperature and barometric pressure of the laboratory were recorded. The atmospheric air density ρ_{atmos} was then calculated from these values using the ideal gas law, as detailed in Section 4.1.

The compressor speed was increased from rest until the outlet pressure drop P_2 reached 2 kPa. At this initial setting, the console readings for P_1, P_2, and P_3 were recorded simultaneously. The compressor speed was then increased in successive steps, each targeting a 1 kPa increment in P_2, with sensor readings recorded at each step. This process was continued until the maximum compressor speed was reached.

At each operating point, the mass flow rate $\dot{m}$ was calculated from P_3 and the ambient conditions using the isentropic inlet flow equation:

$$\dot{m} = a_{\text{inlet}}\sqrt{2\rho_{\text{atmos}}\left(p_{\text{atmos}} - p_{\text{inlet}}\right)} = a_{\text{inlet}}\sqrt{2\rho_{\text{atmos}} \cdot P_3} \tag{3.2.1}$$

where $a_{\text{inlet}} = \pi d_1^2/4$ is the inlet cross-sectional area. Upon completion of data collection, the compressor speed was reduced to zero and the equipment was switched off.

4 Results

4.1 Operating Conditions

The density of air (ρ) can be calculated using the Ideal Gas Law:

$$\rho = \frac{P}{RT} \tag{4.1.1}$$

where:

- P is the absolute pressure (in SI units: Pascals Pa)
- T is the absolute temperature (in Kelvin K)
- R is the specific gas constant for dry air, $R = 287.05\,\mathrm{J\,kg^{-1}\,K^{-1}}$

Given Laboratory Condition Measurements from Table 8.1.1:

$$P = 1025.1\,\mathrm{hPa} = 102\,510\,\mathrm{Pa} \tag{4.1.2}$$

$$T = 22\,\mathrm{C} = 22 + 273.15 = 295.15\,\mathrm{K} \tag{4.1.3}$$

$$R = 287.05\,\frac{\mathrm{J}}{\mathrm{kg\,K}} \tag{4.1.4}$$

$$\rho = \frac{102\,510\,\mathrm{Pa}}{287.05\,\mathrm{J\,kg^{-1}\,K^{-1}}} \times 295.15\,\mathrm{K} \tag{4.1.5}$$

Calculate:

$$\rho = \frac{102510}{287.05 \times 295.15} \tag{4.1.6}$$

$$\approx 1.209\,95\,\frac{\mathrm{kg}}{\mathrm{m^3}} \tag{4.1.7}$$

Thus, **the ambient air density during the experiment is**:

$$\boxed{\rho = 1.21\,\frac{\mathrm{kg}}{\mathrm{m^3}}} \tag{4.1.8}$$

4.2 Measurements

Table 8.1.2 records the raw gauge pressure readings from the three console displays at each compressor speed setting. P_1 and P_2 are given in kPa and P_3 in Pa, reflecting the differing magnitudes of the pressure drops at the throat, outlet, and inlet respectively. The absolute pressures are given below for completeness.

Table 4.2.1: Derived Absolute Presssures

`./outputs/text/pressures.csv`

	p_{atmos} [Pa]	p_{inlet} [Pa]	p_{throat} [Pa]	p_{outlet} [Pa]	$p_{\mathrm{throat}}/p_{\mathrm{atmos}}$ [−]

4.3 Mass Flow Rate

The mass flow rate at each operating point was computed from the inlet gauge pressure $P_3 = p_{\mathrm{atmos}} - p_{\mathrm{inlet}}$ using the isentropic inlet flow equation (Equation 3.2.1):

$$\dot{m} = a_{\mathrm{inlet}}\sqrt{2\rho_{\mathrm{atmos}} \cdot P_3} \tag{4.3.1}$$

where the inlet cross-sectional area is:

$$a_{\text{inlet}} = \frac{\pi d_1^2}{4} = \frac{\pi\,(0.024\,\text{m})^2}{4} \approx 4.524 \times 10^{-4}\,\text{m}^2 \tag{4.3.2}$$

and $\rho_{\text{atmos}} \approx 1.210\,\text{kg/m}^3$ from Section 4.1. The resulting mass flow rates, together with the derived absolute pressures at each operating point, are tabulated in Table 4.3.1 and plotted in Figures 5.1.1 and 5.1.2.

Table 4.3.1: Derived Pressures and Mass Flow Rate at Each Operating Point

`./outputs/text/mass_flow_vs_pressures.csv`

	$p_{\text{atmos}} - p_{\text{outlet}}$ [kPa]	$p_{\text{atmos}} - p_{\text{throat}}$ [kPa]	$p_{\text{atmos}} - p_{\text{inlet}}$ [Pa]	$p_{\text{throat}}/p_{\text{atmos}}$ [−]	$\dot{m}$ [g/s]

5 Analysis and Discussion

5.1 Relations between Mass Flow Rate and Pressures

Figure 5.1.1: Mass Flow Rate $\dot{m}$ vs. Outlet Pressure Drop $p_{\text{atmos}} - p_{\text{outlet}}$

Figure 5.1.2: Throat Pressure Drop $p_{\text{atmos}} - p_{\text{throat}}$ vs. Outlet Pressure Drop $p_{\text{atmos}} - p_{\text{outlet}}$

5.2 Interpretation

Figure 5.1.1 shows $\dot{m}$ rising steeply as the outlet pressure drop $p_{\text{atmos}} - p_{\text{outlet}}$ is first increased from 2 kPa, reflecting the expected isentropic behaviour in the pre-choked regime: a larger pressure differential across the duct drives a higher flow velocity and, consequently, a greater mass flow rate. However, the rate of increase diminishes markedly beyond $p_{\text{atmos}} - p_{\text{outlet}} \approx 6\,\text{kPa}$, and $\dot{m}$ becomes essentially constant at approximately 16.76 g/s for outlet pressure drops of 7 kPa and above. This plateau is the defining signature of choked flow. Figure 5.1.2 corroborates this interpretation. In the pre-choked regime, the throat pressure drop $p_{\text{atmos}} - p_{\text{throat}}$ rises continuously as the outlet pressure is reduced, indicating that pressure disturbances from downstream are still being communicated upstream through the throat. The curve begins to flatten noticeably between $p_{\text{atmos}} - p_{\text{outlet}} = 8\,\text{kPa}$ and 9 kPa, at which point the measured pressure ratio $p_{\text{throat}}/p_{\text{atmos}}$ crosses the theoretical critical value of $r_c = 0.528$ (from 0.546 at 8 kPa to 0.512 at 9 kPa). This crossing marks the onset of choking: the throat Mach number reaches unity, and the throat pressure becomes decoupled from further reductions in p_{outlet}. Accordingly, **choking first occurs at approximately $p_{\text{atmos}} - p_{\text{outlet}} \approx 8\,\text{kPa}$ to 9 kPa.**

5.3 Comparison to Theory

Table 5.3.1 summarises the comparison between the experimentally obtained limiting values and the isentropic theoretical predictions.

Table 5.3.1: Comparison of Experimental and Theoretical Limiting Values

Quantity	Experimental	Theoretical	Relative Error
$\dot{m}_{\text{max}}$ [g/s]	16.757	17.093	1.97%
$p_{\text{throat}}/p_{\text{atmos}}$ at choking	0.481	0.528	8.90%

The experimental $\dot{m}_{\max}$ agrees closely with theory, with a relative error of under 2%. This small discrepancy is consistent with the assumptions of the isentropic model: in practice, viscous boundary layers and minor frictional losses in the duct reduce the effective flow capacity marginally below the inviscid prediction.
The limiting throat pressure ratio shows a larger deviation of approximately 8.9%, with the experimental value of 0.481 falling below the theoretical $r_c = 0.528$. This is also physically consistent with real-flow effects: friction and boundary layer growth in the convergent section act to reduce the effective throat area, requiring a lower throat pressure (i.e. a higher pressure drop) to drive the flow to Mach 1 compared to the frictionless ideal. Additionally, the discrete 1 kPa step resolution of the compressor speed control means the exact choking point cannot be pinpointed with greater precision than the interval between consecutive readings.

6 Summary and Conclusions

This experiment investigated the phenomenon of choked flow in a convergent-divergent duct by measuring the inlet, throat, and outlet pressures across a range of compressor speeds. The key findings are as follows:

1. **Choking was clearly observed.** The mass flow rate $\dot{m}$ increased steadily with outlet pressure drop up to approximately 7 kPa, beyond which it plateaued at a constant value of 16.76 g/s, consistent with the theoretical prediction that $\dot{m}$ reaches a maximum when $M_{\text{throat}} = 1$. The onset of choking was identified between outlet pressure drops of 8 kPa and 9 kPa, where the throat pressure ratio $p_{\text{throat}}/p_{\text{atmos}}$ crossed the critical value r_c.

2. **The experimental $\dot{m}_{\max}$ agrees closely with theory.** The measured limiting mass flow rate of 16.76 g/s falls within 2% of the isentropic prediction of 17.09 g/s, confirming that the isentropic model is a good approximation for this flow configuration despite the simplifying assumptions of frictionless, adiabatic flow.

3. **The limiting throat pressure ratio deviates more notably from theory.** The experimental value of $p_{\text{throat}}/p_{\text{atmos}} = 0.481$ lies 8.9% below the theoretical $r_c = 0.528$. This is attributed to viscous and boundary layer effects in the convergent section, which reduce the effective throat area and require a greater pressure drop to accelerate the flow to sonic conditions, as well as the 1 kPa resolution limit of the pressure measurements.

Overall, the experiment successfully demonstrated the choking phenomenon and validated the core predictions of isentropic compressible flow theory. The close agreement in $\dot{m}$ and the physically interpretable discrepancy in r together support the use of the isentropic model as a practical first-order tool for analysing convergent-divergent duct flows, with the understanding that real-flow corrections may be necessary for higher-fidelity predictions.

7 References

[1] J. D. Anderson and C. P. Cadou, *Fundamentals of Aerodynamics*, 7th ed. McGraw-Hill Education, March 17, 2023, ISBN: 9781266076442.

8 Appendices

8.1 Data

Table 8.1.1: Ambient Laboratory Conditions during Experiment

Quantity	Value	Uncertainty
Lab. Temperature, T	22 °C	±0.5 °C
Lab. Pressure, P	1025.1 hPa	±0.05 hPa

Table 8.1.2: Raw Measurements of Pressure Taps

`./data/pressure_taps.csv`

P_1 [kPa]	P_2 [kPa]	P_3 [Pa]

8.2 Code

For the sake of brevity, only the code files that are key to the analysis are included below. However, in the spirit of completeness, the repository containing the complete data, source code, and notes for this report can be found at github:vaisriv/enae464-lab04.

Listing 8.2.1: Index File

`./src/index.py`

```python
# FIXME:
# scaffold enae464-lab04

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.ticker as ticker
from pathlib import Path

# ── Paths ──────────────────────────────────────────────────────────────

DATA_DIR        = Path("./data")
OUTPUT_TEXT_DIR = Path("./outputs/text")
OUTPUT_FIGS_DIR = Path("./outputs/figures")

OUTPUT_TEXT_DIR.mkdir(parents=True, exist_ok=True)
OUTPUT_FIGS_DIR.mkdir(parents=True, exist_ok=True)

# ── Constants ──────────────────────────────────────────────────────────

P_ATMOS   = 102510.0   # Pa  (1025.1 hPa from Table 1)
T_ATMOS   = 295.15     # K   (22°C from Table 1)
R_AIR     = 287.05     # J/(kg·K)
RHO_ATMOS = P_ATMOS / (R_AIR * T_ATMOS)   # ≈ 1.210 kg/m³

```

```python
GAMMA   = 1.4
D1      = 0.024        # m  (inlet diameter)
D2      = 0.0095       # m  (throat diameter)
A_INLET = np.pi * D1**2 / 4
A_THROAT = np.pi * D2**2 / 4

# ── Load raw data ──────────────────────────────────────────────────────
# Columns as recorded from the C1-MKII console:
#   P1 [kPa]  =  p_atmos - p_throat
#   P2 [kPa]  =  p_atmos - p_outlet
#   P3 [Pa]   =  p_atmos - p_inlet

raw = pd.read_csv(DATA_DIR / "pressure_taps.csv",
                  names=["P1_kPa", "P2_kPa", "P3_Pa"],
                  header=0)

# Convert gauge readings to absolute pressures [Pa]
p_throat = P_ATMOS - raw["P1_kPa"] * 1e3
p_outlet = P_ATMOS - raw["P2_kPa"] * 1e3
p_inlet  = P_ATMOS - raw["P3_Pa"]

# ── Build relative-pressure table ──────────────────────────────────────

rel = pd.DataFrame({
    "p_atmos [Pa]":           P_ATMOS,
    "p_inlet [Pa]":           p_inlet.round(4),
    "p_throat [Pa]":          p_throat.round(4),
    "p_outlet [Pa]":          p_outlet.round(4),
    "p_throat / p_atmos [-]": (p_throat / P_ATMOS).round(6),
})

rel.to_csv(OUTPUT_TEXT_DIR / "pressures.csv", index=False)
print(rel.to_string(index=False))
print(f"\nSaved → ./{OUTPUT_TEXT_DIR / 'pressures.csv'}")

# ── Mass flow rate ─────────────────────────────────────────────────────
# From the isentropic inlet flow equation:
#   ṁ = a_inlet · √( 2 · p_atmos · (p_atmos - p_inlet) )
#     = a_inlet · √( 2 · p_atmos · P3 )

mdot = A_INLET * np.sqrt(2 * RHO_ATMOS * raw["P3_Pa"])

# ── Build mass flow vs. pressures table ────────────────────────────────

dp_outlet = raw["P2_kPa"] * 1e3   # p_atmos - p_outlet [Pa]
dp_throat = raw["P1_kPa"] * 1e3   # p_atmos - p_throat [Pa]

mvp = pd.DataFrame({
```

```python
    "p_atmos - p_outlet [Pa]": dp_outlet.round(4),
    "p_atmos - p_throat [Pa]": dp_throat.round(4),
    "p_atmos - p_inlet [Pa]":  raw["P3_Pa"].round(4),
    "p_throat / p_atmos [-]":  (p_throat / P_ATMOS).round(6),
    "mdot [kg/s]":             mdot.round(6),
})

mvp.to_csv(OUTPUT_TEXT_DIR / "mass_flow_vs_pressures.csv", index=False)
print(f"\n{mvp.to_string(index=False)}")
print(f"\nSaved → ./{OUTPUT_TEXT_DIR / 'mass_flow_vs_pressures.csv'}")

# ── Theoretical choked limits ──────────────────────────────────────────
# Critical pressure ratio: r_c = (2 / (γ+1))^(γ/(γ-1))
# Choked ṁ via the throat form of the isentropic mass flow equation

r_c       = (2 / (GAMMA + 1)) ** (GAMMA / (GAMMA - 1))
mdot_max  = (RHO_ATMOS * A_THROAT
             * np.sqrt(GAMMA * (P_ATMOS / RHO_ATMOS)
                       * (2 / (GAMMA + 1)) ** ((GAMMA + 1) / (GAMMA - 1))))
dp_throat_choked = (1 - r_c) * P_ATMOS   # p_atmos - p_throat at choking

# ── Plot 1: ṁ vs. (p_atmos - p_outlet) ────────────────────────────────

fig, ax = plt.subplots(figsize=(7, 4.5))

ax.plot(dp_outlet / 1e3, mdot * 1e3,
        marker="o", linewidth=1.5, markersize=5,
        color="steelblue", label="Experimental")

ax.axhline(mdot_max * 1e3,
           color="tomato", linewidth=1.5, linestyle="--",
           label=rf"Theoretical $\dot{{m}}_\mathrm{{max}}$ = {mdot_max*1e3:.3f} g/s")

ax.set_xlabel(r"$p_\mathrm{atmos} - p_\mathrm{outlet}$ [kPa]")
ax.set_ylabel(r"$\dot{m}$ [g/s]")
ax.set_title(r"Mass Flow Rate vs. Outlet Pressure Drop")
ax.xaxis.set_minor_locator(ticker.AutoMinorLocator())
ax.yaxis.set_minor_locator(ticker.AutoMinorLocator())
ax.grid(which="major", linestyle="--", alpha=0.4)
ax.grid(which="minor", linestyle=":",  alpha=0.2)
ax.legend()
fig.tight_layout()
fig.savefig(OUTPUT_FIGS_DIR / "mdot_vs_dp_outlet.png", dpi=200)
plt.close(fig)
print(f"\nSaved → ./{OUTPUT_FIGS_DIR / 'mdot_vs_dp_outlet.png'}")

# ── Plot 2: (p_atmos - p_throat) vs. (p_atmos - p_outlet) ──────────────

```

```python
fig, ax = plt.subplots(figsize=(7, 4.5))

ax.plot(dp_outlet / 1e3, dp_throat / 1e3,
        marker="o", linewidth=1.5, markersize=5,
        color="steelblue", label="Experimental")

ax.axhline(dp_throat_choked / 1e3,
           color="tomato", linewidth=1.5, linestyle="--",
           label=rf"Theoretical choked throat $\Delta p$ = {dp_throat_choked/1e3:.3f} kPa"
                 rf"  $(r_c = {r_c:.3f})$")

ax.set_xlabel(r"$p_\mathrm{atmos} - p_\mathrm{outlet}$ [kPa]")
ax.set_ylabel(r"$p_\mathrm{atmos} - p_\mathrm{throat}$ [kPa]")
ax.set_title(r"Throat Pressure Drop vs. Outlet Pressure Drop")
ax.xaxis.set_minor_locator(ticker.AutoMinorLocator())
ax.yaxis.set_minor_locator(ticker.AutoMinorLocator())
ax.grid(which="major", linestyle="--", alpha=0.4)
ax.grid(which="minor", linestyle=":",  alpha=0.2)
ax.legend()
fig.tight_layout()
fig.savefig(OUTPUT_FIGS_DIR / "dp_throat_vs_dp_outlet.png", dpi=200)
plt.close(fig)
print(f"\nSaved → ./{OUTPUT_FIGS_DIR / 'dp_throat_vs_dp_outlet.png'}")
```